

I. INTRODUCTION

These supporting materials accompany the research paper *Extract, then decide: a two-stage pipeline to identify seizure frequency patterns in epilepsy clinic letters*.

It is composed of four sections: an extended literature review; a description of the project life cycle (planning, development, implementation, testing, and evaluation); descriptions of the tools used to build and run the system; and a critical appraisal of design decisions, lessons learned, and the outcome with hindsight. Three appendices follow the references: the extract prompt, the decide prompt, and a web app demonstration.

II. EXTENDED LITERATURE REVIEW

A. Complexity of clinical narratives

Clinical letters use narrative language so that clinicians can record the patient’s situation, their judgement, and their uncertainty. That flexibility improves clinical communication, but it makes the information difficult to use at scale [1]. Clinicians often use hedging to describe certainty, frequency, and quantity rather than stating a precise value [2]. The task of information extraction (IE) is to convert that nuanced information into a standardised form while preserving its essential meaning.

Epilepsy clinic letters illustrate that complexity. One letter can contain diagnoses, prescriptions, history, seizure descriptions, and several references to seizure frequency. Fonferko-Shadrach *et al.* report much lower human agreement for seizure frequency (per-item F1 = 0.47) than for prescriptions (0.87) or diagnoses (0.83) [3]. They attribute this to the variety of frequency expressions and to mentions of more than one seizure or event type. Effective extraction requires clearly defined policies and procedures describing what to extract and how to represent it.

B. Seizure frequency as a selection task

The same letter can be used for different purposes. ExECT and the later annotation study treat it as an inventory: they record several entity mentions and their attributes. One letter may list several anti-seizure medications, each with a name, dose, unit, and administration frequency [3], [4]. That form is appropriate when the aim is building a broad clinical inventory.

The task in this study is different. Gan *et al.* assign each letter one label for the patient’s current seizure frequency status [5]. The system has to gather candidate descriptions, interpret their meaning and time frame, and choose the single normalised label that best represents the current pattern. Earlier one-label studies use the same formulation [6]–[8]. The single label extraction implies that it is all one task, but this study proposes two separate stages: extracting every candidate with its evidence in a canonical form, then deciding the final answer from those candidates.

Two common cases show why that split matters. Diary-style counts may need to be combined: *1 in June, 0 in July, 2 in August* can become *3 in 3 months*. Competing statements may instead require precedence: *She may remain seizure-free for up to 4 months, but then will experience clusters of 5 seizures in a single day*. Different annotation guidelines would

TABLE I: Past seizure frequency papers: headline results

Author	Year	Method	F1 score
Xie <i>et al.</i>	2022	Fine-tuned BERT	0.83
Decker <i>et al.</i>	2022	Rules	0.82
Fonferko-Shadrach <i>et al.</i>	2024	Rules	0.66
Fernandes <i>et al.</i>	2024	Fine-tuned BERT	0.37
Holgate <i>et al.</i>	2024	Prompted LLM	0.73
Holgate <i>et al.</i>	2025	Fine-tuned LLM	0.68
Gan <i>et al.</i>	2026	Fine-tuned LLM	0.79
Brown & Gan	2026	Hybrid	0.80

record those cases differently. Separate stages make the policy choice inspectable rather than implicit.

C. Rules, BERT models, and modern LLMs

Several papers have published seizure frequency extraction systems using rules, fine-tuned models, or prompted large language models. Rules match terms and apply domain logic. ExECT does this for a clinical inventory; Decker *et al.* use regular expressions for seizure-related variables [4], [9]. Fine-tuned BERT models learn the mapping from labelled examples, as in Xie *et al.* and Fernandes *et al.* [6], [10]. Table I lists the F1 scores those studies report. The scores are not a ranking of methods: each study uses its own dataset, label set, and evaluation.

Prompted and fine-tuned LLMs are more recent. Holgate *et al.* show that a prompted model can extract seizure information from instructions and examples [7]. Holgate *et al.* and Gan *et al.* later fine-tune models to a defined output format [5], [8].

BERT-style encoders and decoder-based LLMs share the transformer architecture. The practical difference is in-context learning: an LLM can follow instructions and examples without task-specific training [11]. That behaviour transfers to clinical extraction, where clear instructions and examples can reduce the need for heavy post-processing [12]. Gan *et al.* show that combining structured prompting and fine-tuning can deliver even stronger performance. This study explores whether structured prompting in a two-stage pipeline can deliver comparable performance to a fine-tuned LLM.

D. Modular extraction pipelines

Multi-stage clinical extraction pipelines split a task into distinct stages instead of asking a model for one final answer in a single step. CLINES uses stages for chunking, extraction, attribute assignment, normalisation, date resolution, and aggregation [13]. Seizure frequency studies have also decomposed the task in structured prompting including numbered reasoning steps and summarise-then-extract designs [7], [14].

This study decomposes the task into two stages: extracting candidate evidence into a structured record, and deciding the final answer from that record. One LLM call performs the extraction. The decision stage can be performed by rules or a second LLM call, so the two methods can be compared with the extraction held fixed. Past research on seizure frequency gives rules a minor repair role after an LLM; this study

evaluates whether they can play a more substantial, explicit role in the pipeline.

III. PLANNING

The project lifecycle is divided into five stages: planning, development, implementation, testing and evaluation. The sections below describe the key decisions, the rationale behind them, and the sources they were based on.

There are two key documents in the planning phase:

- 1) The project brief, created by my supervisor.
- 2) The problem specification and project plan, created by me.

A. Project brief

The project brief outlines the initial scope of the project. The problem specification and project plan refines the scope, specifies the research objectives and describes the methods used to achieve them. The project brief was:

Instead of training models, this project builds a training-free (or minimal-training) multi-agent extraction system that reads epilepsy letters and outputs structured fields (e.g., ASM, seizure type/frequency, investigations, epilepsy syndrome/type). The student will design roles such as: (a) Section/Time-line Agent (segments text, builds timeline), (b) Field Extractor Agents (one per field group), (c) Verification Agent (checks evidence spans, contradictions, missingness), and (d) Aggregator Agent (produces final JSON + confidence + citations to text spans). The key research goal is reliability: compare single-prompt extraction vs multi-agent extraction under the same budget constraints, and test improvements from self-consistency, evidence requirements (“answer only if supported by quote”), and structured output validation. Evaluation uses field-level accuracy/F1, and robustness tests. Data can be synthetic or de-identified samples if available. Deliverables: an end-to-end agentic pipeline, evaluation harness, and a dissertation focused on engineering + empirical study of reliability.

The deliverables are a narrow specification of what a successful project would look like:

- 1) An end-to-end agentic pipeline.
- 2) An evaluation harness.
- 3) A dissertation focused on engineering + empirical study of reliability.

The type of data this would be applied to was narrowly defined as epilepsy letters, but it was undetermined whether the specific data source was synthetic or de-identified samples. This was one key decision to be agreed in the project kick-off meeting.

The research would compare a single-prompt extraction vs. multi-agent extraction, and the multi-agent system included a number of initial suggestions such as verification agent and aggregator agent. The precise implementation of the multi-agent system would be defined as the research progresses,

guided by best practices in the literature and the results from empirical experiments.

In the initial project kick-off meeting it was agreed that the specific data source used would be a set of 1,500 synthetic clinical letters that the supervisor had access to. As these were synthetic letters, there would be no need to get specific ethics approval to use for research purposes.

B. Problem Specification and Project Plan

The problem specification and project plan was uploaded to the university Canvas site on the 5th June 2026. This is a direct extract from the methods section:

There are two primary data sources in this project:

- 1,500 synthetic clinical letters and structured labels from Gan *et al.* [5]
- 200 synthetic clinical letters and structured labels from Fonferko-Shadrach *et al.* [3]

The project will use LLMs as the core AI tool, using both open (e.g. Qwen 3.6) and closed (e.g. GPT 5.4) models. The research will involve building an agent harness around the model that enables it to ingest clinical letters and prompt instructions, produce a structured output, and have that output validated and evaluated.

The research will be empirical in nature and focus primarily on assessing the quality, reliability and validity of the outputs. Cost, latency, data schema complexity and evidence specificity will be key experimental variables tested.

Evaluation will take place in three key forms: comparing the accuracy (F1 score) of the outputs vs. the ground-truth labels from the two data sources, assessing the validity of the structured data schema created, and assessing the validity of the evidence cited. A series of ablations will be conducted to isolate the impact of different components of the agent harness.

The F1 score evaluated on a set of ground-truth labels is the de-facto standard for evaluating AI models on how well they can complete a classification task. Evaluating the model on two different datasets optimised for two different tasks effectively tests the flexibility of the system.

The ExECTv2 dataset [3] was used extensively as per the original plan. The pipeline was built, experiments were conducted and results were analysed in depth and benchmarked against the prior results. However, when the time came to write the paper it became clear that writing about two datasets with different annotation policies, evaluation measures, error profiles and sample sizes required the reader to hold too much information in mind.

As a result, the empirical analysis in the paper and the primary evaluation in these materials focus on the seizure-frequency extraction task on the Gan letters. ExECTv2 appears here only as design history and as the descriptive

inventory panel in Section VII-G. We used the pipeline built for ExECTv2 to extract diagnoses, medications and investigations from the main dataset of 1,500 synthetic letters as that feasibility study; some of the design choices for the seizure-frequency extraction pipeline are directly informed by designing a modular system that can be applied to both tasks. Results benchmarked against the ExECTv2 dataset will be published in a separate paper.

The following sections describe how the rest of this plan was implemented.

IV. DEVELOPMENT

This section explains how the initial project proposal was translated into a focused and testable research approach. It records the evidence, design choices, and trade-offs that led to the final method.

A. Original design requirements

The original brief contained a series of key design requirements for the pipeline. It must:

- 1) Ingest epilepsy letters
- 2) Output validated structured fields
- 3) Contain verified evidence citations
- 4) Produce this in an aggregated JSON

These design requirements were tightly coupled to a multi-agent framework, e.g. using a *Verification Agent* to check evidence spans and an *Aggregator Agent* to produce the final JSON and citations.

The literature review showed that multi-agent systems are particularly useful for complex tasks [15], especially on long-context tasks [16]. However, research also shows that single-agent systems are often as effective when they are given equal compute and multi-agent systems create new failure modes as errors propagate throughout the system [16].

For this task, the full epilepsy letter and instructions fit comfortably in the LLM’s context window so there is no need for convoluted multi-hop pipelines or document splitting that multi-agent systems performed particularly well on [13]. Thus, a multi-agent system cannot be justified on that basis.

It is less clear whether this task is sufficiently complex to justify a multi-agent system. Past research has shown that LLMs struggle with temporal logic and numerical logic, and both occur in this task. It is also a task that human annotators struggle to do reliably and consistently [3]. This suggests the task is at least somewhat complex and a multi-agent system is justified on that basis.

However, it is important that each LLM call is given a sufficiently complex task to justify the added system complexity. A key research objective is to design a reliable system. A reliable system is one that contains the minimum amount of complexity necessary to function. More components create more failure modes, and in a sequential process those errors propagate and multiply at each step.

B. Deterministic code for deterministic tasks

A key design choice early on was to simplify. Any task that can be reliably completed by a deterministic system (code) instead of a probabilistic system (LLMs) should be done deterministically.

Evidence verification and schema validation are two tasks that code can do reliably and efficiently. Initial experiments showed that LLMs cannot do this task reliably; they introduce as many new schema errors as they repair, and they occasionally miss invalid evidence. Even if they could do the task reliably, code already does it at a lower cost and latency.

Therefore a dedicated *Aggregator Agent* to produce the final JSON with text citations and a dedicated *Verification Agent* to check evidence spans were removed from the design after initial experiments. A *Field Extractor* agent was no longer relevant as the task was limited to a single field, seizure frequency. Initial experiments showed that a *Section/Timeline Agent* was a net-negative contribution because correct seizure frequency references were not always in the dedicated seizure frequency section, and splitting up the text encouraged the LLM to select wrong answers more frequently.

C. Pipeline design review

After initial experiments, all four of the original proposed LLM agents had been ruled out because they were now either invalid or empirically performed worse than a single LLM. This led to a major review of the pipeline design.

Comprehensive error analysis showed that the LLM was consistently finding valid seizure frequency patterns, providing exact evidence from the text and often choosing defensible answers based on the instructions. It did not need another LLM call to check its evidence or structure it into JSON; it needed cleaner task separation.

The work separates two clinical operations: extracting every relevant mention of seizure frequency from the letter, already written in the form the gold-standard labels expect, and applying a policy to choose which label matches the gold-standard answer. The labelling conventions and policy choices in the gold-standard evidence are valid, but they are not the only valid choice. Therefore a key pipeline requirement is producing results that conform to an explicit set of subjective choices.

The original brief separated mechanical tasks across multiple LLM calls; one call splits the evidence, another checks the evidence, another aggregates it together. With simplicity and efficiency in mind, it would be preferable if a single LLM call could do all of those tasks at once. This would require the LLM to generate a much richer intermediate schema. Rather than extracting a single seizure frequency label, the LLM would extract every relevant seizure frequency pattern as a distinct entity in a nested JSON, with exact evidence for every entity, and make an initial proposal.

Using this richer schema, a second LLM call could then review the evidence and apply the correct policy in hard cases. This reframes the decomposition around clinical tasks (extract the candidates, then decide between them) instead of

mechanical tasks. The richer record also makes it possible for deterministic rules to apply the same policy to the evidence the LLM extracted.

Past research shows that deterministic code can reliably normalise seizure frequency values, and hybrid LLM and rules systems can outperform LLMs alone. Rules-based systems struggled with seizure frequency extraction more than medications, but they still performed better than human experts. Deterministic code already performed schema validation and evidence verification more reliably than LLMs, so there was a case for more deterministic code in the pipeline.

D. Final pipeline design

The final outcome of this design review was a pipeline with two stages, *extract* and *decide*. *Extract* finds all seizure frequency patterns in the letter, records each one with a normalised label and the exact quote supporting it, and selects the one that best represents the patient’s seizure frequency pattern with a one sentence rationale. *Decide* reviews that evidence against the established policy and decides whether to keep, change or revise the provisional answer.

This study’s core comparison holds the extraction call fixed and changes only who performs *decide*: rules (*Hybrid*) or a second LLM call (*LLM-only*).

The original project brief required a pipeline that could ingest epilepsy letters, output validated structured fields, contain verified evidence citations and produce this in an aggregated JSON. It required an evaluation harness and a dissertation focused on engineering and an empirical study of reliability. The following sections demonstrate how this was implemented.

V. IMPLEMENTATION

The implementation was designed to produce a structured, reproducible account of current seizure frequency from a clinical letter, while retaining the evidence and intermediate choices needed to inspect how an answer was reached. It describes the reusable pipeline and the controlled comparison design; performance results and the findings of individual ablations are reported in the evaluation section.

A. Structured record

The pipeline receives a synthetic clinical letter as input and produces a structured seizure frequency record. The final record contains the submitted frequency label, direct supporting evidence, and the metadata needed to identify the method, model configuration, data split, and scoring policy used. This design makes the final answer usable for benchmark evaluation without reducing the whole process to an opaque single-label extraction.

The pipeline uses a richer intermediate record than the final benchmark label. During extraction, it can retain multiple candidate seizure frequency events, their source spans, raw wording, time information, and an initial proposed answer. This is important because a letter may contain several plausible statements: historical and current rates, periods without seizures, isolated breakthrough events, clusters, and uncertain

or qualitative descriptions. Preserving the alternatives avoids losing information before the annotation policy has been applied.

B. Extract and decide

The record distinguishes the source wording from the canonical label form and from the final answer. The *decide* stage can revise the provisional answer while retaining the evidence on which the original proposal was based. The final output is validated against the expected schema, while source data and raw model output are retained separately from format repair and clinical policy decisions.

The interface between the two stages is fixed. *Extract* receives the full letter and returns the candidate record: every seizure frequency event with its quoted span, category and canonical label, plus a provisional answer. *Decide* receives that record and the decision policy, and returns the final label with the event(s) that support it. In *Hybrid*, the decision rules also rewrite any label the model left in a non-canonical form.

C. Rules and deterministic components

Deterministic components implement tasks for which explicit, reproducible handling is appropriate. These include parsing and schema repair, validation of required evidence fields, aggregation of the final record, and rules for normalising frequency expressions into the task’s closed vocabulary. The implementation keeps schema repair, benchmark representation, and clinically meaningful policy rules distinct and are evaluated separately.

In *Hybrid*, recorded rules perform the decision role. Their authority is recorded by rule type: whether a rule merely standardises a label form (dialect or form), rewrites the submitted label from the extracted evidence (e.g. totalling a seizure diary), or chooses a different already-extracted event. No decision rule scans the letter for a new candidate. This makes deterministic processing an explicit part of the method rather than unreported post-processing.

D. LLM Components

The LLM implementation uses structured prompts that ask the model to return a record containing candidate seizure-frequency events, evidence, and a proposed answer. The extraction call reads the letter and returns that record. In *LLM-only*, a second call receives the record and adjudicates between competing candidates under the same policy that the rules apply, written as instructions with worked examples.

Appendix A is the *extract* template: task, instructions, allowed label forms, event schema and decision schema. The four components ablated in the accompanying paper map onto it as follows: the *instructions* are the task and instruction blocks; the *examples* are the worked examples inside the instructions; the *allowed labels* are the closed `label_forms` block; and the *evidence obligation* is the evidence fields of the event schema together with the exact-quote instruction. The letter is attached at run time as `note_text` and is omitted there. Appendix B is the *decide* prompt used by *LLM-only*:

TABLE II: Seizure frequency label forms

Family	Written forms
Rate	N per unit; N per N unit; N per N to N unit; N to N per unit; N to N per N unit; multiple per unit; multiple per N unit; N per multiple unit
Cluster	cluster per unit, N per cluster; cluster per unit, range per cluster; cluster per unit, multiple per cluster
Seizure free	seizure free for a duration
Unknown	unknown; no seizure frequency reference

TABLE III: Seizure frequency event categories in the extraction record

Category	Use
frequency_rate	A countable or range rate
cluster_frequency	How often clusters occur
seizure_free	A seizure-free statement
last_event_only	A dated last event without a rate
unknown_frequency	Seizures discussed, no usable rate
no_reference	No usable frequency evidence

TABLE IV: Decision stage examples

Events	Provisional answer	After decide
3 in March; 6 in May	6 per month	9 per 3 month
Last seizure on 12th March; well 3 weeks	seizure free for 3 week	1 per month

the same policy written as instructions with worked examples, plus the allowed label forms and the decision schema. The evidence record from `extract` is attached at run time as `first_choice` and `events`; the letter is not. Event categories are listed in Table III, label forms in Table II, and short decide examples in Table IV.

VI. TESTING

A test suite was built to check that letters and gold labels loaded without silent changes, that development and held-out letters stayed apart, and that work on prompts and rules used only the development set. Held-out results were reported only as totals. The same suite kept a short list of known hard cases: adding up a seizure diary, not counting the same evidence twice, handling empty months, reading clusters, and preferring the current rate over an older one. When a prompt or rule changed, those cases were re-run so a local fix did not break earlier behaviour. End-to-end checks confirmed that `extract` then `decide` still wrote a valid record.

After the package was released, my supervisor installed it on the university high-performance computing cluster and ran both methods on a new dataset. Both runs finished and wrote structured output. That shows the package can be installed and used in other environments.

VII. EVALUATION

This section evaluates whether the verified pipeline met the research objective: to extract a patient’s current seizure frequency burden from a clinical letter reliably, with a structured evidence record and a reproducible comparison between

TABLE V: Purist micro-F1 on synthetic and real clinic letters

Method	Synthetic dev	Synthetic test	Real letters
Hybrid	0.83	0.82	0.80
LLM-only	0.77	0.77	0.77

rules and LLMs. The accompanying paper reports the headline held-out comparison: on one shared Gemini extract record, `Hybrid` reaches 0.86 Purist and `LLM-only` 0.85. This section expands contract adherence, prompt-mechanism detail, secondary decompositions, generalisation, and evidence-quality limits.

A. Independent real-letter comparison

DeepSeek V4 Flash was evaluated on the set of 300 clinic letters using real patient data [5] and demonstrated the method transfers and result patterns hold. `Hybrid` remained ahead of `LLM-only` (0.80 vs. 0.77) and matched the best published synthetic-trained fine-tune on the same letters (0.79) [5]. Results are shown in Table V.

B. Extracting multiple candidates with exact evidence

A key feature of the pipeline is the rich extraction record. The model is asked to extract every seizure frequency pattern in the text, quote an exact evidence span for each, write each in a canonical form, and propose which best represents the patient’s current status. That record is what allows the second LLM call in `LLM-only` and the rules in `Hybrid` to re-use the evidence and improve on the provisional answer by +0.06 to +0.07. The cost is a more complex schema, and this is where the local models fall slightly behind.

The gold standard records one label and one quote per letter. The extraction record holds more: on `test450`, Gemini extracted 985 events over 450 letters (2.19 per letter) and the six models ranged from 2.19 to 2.82 events per letter. The record therefore preserves alternatives that a one-label output would discard.

Table VI shows each model’s adherence to the extraction schema on `test450`. The four hosted models quoted an exact substring of the letter for the selected event in 99% or more of letters; Qwen (97%) and Gemma (93%) occasionally paraphrased. Gemma also needed the most non-semantic format repair (15 letters), typically doubled braces or quotation marks that broke the nested JSON, and left 8 letters unparsed after repair; those letters are scored as incorrect. Format repair is recorded separately from every semantic rule and never changes a label.

C. Development vs. test results

Figure 1 shows the two methods on both splits. `Hybrid` scores 0.87 on development and 0.86 on the held-out set, a generalisation gap of -0.01 . `LLM-only` scores 0.85 on both. Neither method was tuned on held-out letters; prompts and rules were fixed on the development split and replayed once.

TABLE VI: Six models as the extraction call on test450: scores and contract adherence (Hybrid)

Model	Purist	Prag.	Exact evidence	Repairs	Unparsed	Retry rej.	Events
Gemini 3.7 Flash	0.86	0.88	99.8% (449)	0	1	0	985
Grok 4.6	0.85	0.89	100% (450)	0	0	0	1063
DeepSeek V4 Flash	0.82	0.85	99.6% (448)	2	0	0	1267
GPT-5.6 Luna	0.79	0.82	98.9% (445)	0	1	0	1102
Qwen 3.8 27B	0.76	0.81	96.7% (435)	3	2	0	1004
Gemma 4 26B	0.72	0.77	92.9% (418)	15	8	7	1068

Exact ev.: the quoted span for the selected event is an exact substring of the letter. Repairs: letters needing a non-semantic JSON dialect repair. Unparsed: no valid record after repair. Retry rej.: format-only retries whose output was rejected. Events: extracted events over 450 letters. Source: `paper_experiments/gan/runs/{model}/test450/comparison.json`.

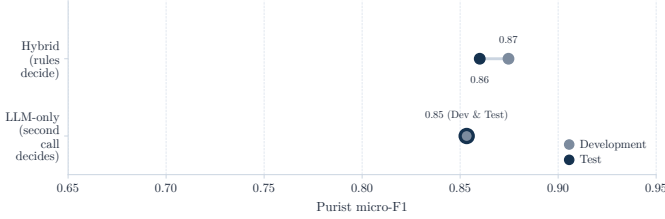


Fig. 1: The two methods on the development and held-out splits, Purist micro-F1. Both share the same extraction call. Hybrid moves from 0.87 to 0.86; LLM-only is 0.85 on both.

D. Error examples

The accompanying paper’s error analysis concentrates on incorrect unknown answers. The figures below are illustrations of that of those types of errors (seizure-free, frequent, and infrequent gold labels each submitted as unknown). They demonstrate how the model reasons under uncertainty.

E. Healthcare capability index

Figure 5 plots each model’s Hybrid Purist micro-F1 on test450 against a published healthcare capability index for that model. Medical intelligence is closely correlated with performance in this clinical extraction task.

F. Experimental environment

Experiments using local language models took place on a Dell XPS 16 9640 running Windows 11, with an Intel Core Ultra 9 185H, 32 GB RAM, and an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB VRAM). Ollama v0.32.15 was used to serve the two local models (Qwen 3.8 27B, Gemma 4 26B). Those local results show that the Hybrid stack can run on a single laptop GPU under the same synthetic task conditions; they are not real-letter performance, clinical validity, or deployment readiness.

Experiments using hosted APIs were run from a Mac mini M1 with 16 GB unified memory on macOS 26.5.2. API calls were made directly to the provider (OpenAI, DeepSeek) or through model routers (OpenRouter, AI Gateway). Four hosted models were used (Gemini 3.7 Flash, GPT-5.6 Luna, Grok 4.6, DeepSeek V4 Flash). Hosted accelerators are not disclosed by the providers.

All models used a temperature of 0 except Luna, which only accepts a temperature of 1. An ablation was run on Gemini using a temperature of 1. All models used a thinking or effort level of low. Ablations were run on Gemini using thinking levels of medium and high.

Experiments were run with Python 3.14.4. DSPy 3.2.1, Pydantic 2.13.4 and LiteLLM 1.87.1 were the core packages. DSPy co-ordinated the model calls and required structured JSON output. Pydantic validated that the structured output conformed to the schema. LiteLLM delivered API calls to Ollama and the model routers. Every reported cell is replayable from saved model output in the repository’s `paper_experiments/` directory without a new model call.

G. Descriptive clinical inventory feasibility

This task focused on one current seizure-frequency label per letter. The same synthetic letters also mention diagnoses, medicines and investigations. To show that broader structured content, the inventory schema and extraction program built for the separate ExECTv2 task [3] was applied to a sample of 100 letters in this dataset. Summary results are in Table VII. An example of an annotated letter is shown in Figure 6.

VIII. TOOLS

A. Development environment and implementation

The implementation used Python (version 3.11 or later), which provided one language for data loading, deterministic extraction rules, experiment orchestration, and evaluation. DSPy supplied the structured interface for language-model programs and their typed inputs and outputs. LiteLLM provided a common model-access layer across hosted providers and the local Ollama route, allowing the experiment code to retain the same orchestration pattern while changing the model endpoint. Pydantic was used for explicit data models and validation, helping keep intermediate records and experiment configuration machine-readable.

B. Model access and inference infrastructure

Hosted models were accessed through provider APIs, with DSPy and LiteLLM handling the shared calling interface, request settings, timeouts, and retry behaviour. This made it possible to compare several providers within the same experiment design without treating their serving hardware as part of the local development environment.

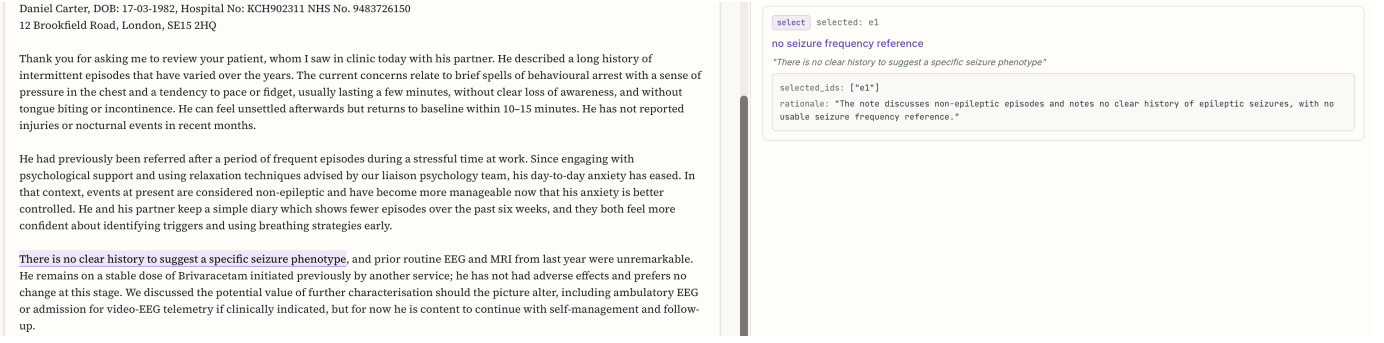


Fig. 2: Seizure free → unknown (development example). Gold-standard label: seizure free for multiple months. Model selected no seizure frequency reference (unknown).

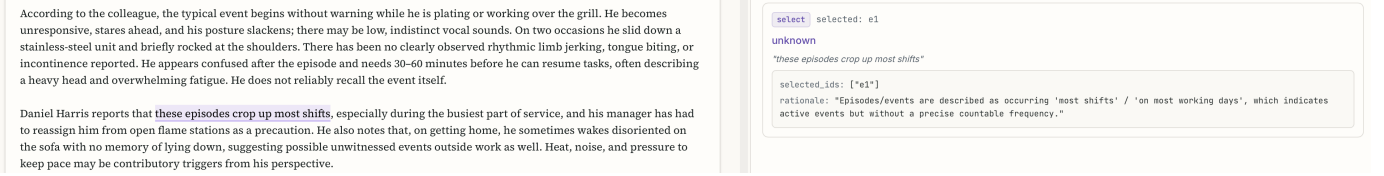


Fig. 3: Frequent → unknown (development example). Gold-standard label: multiple per week. Model selected unknown.

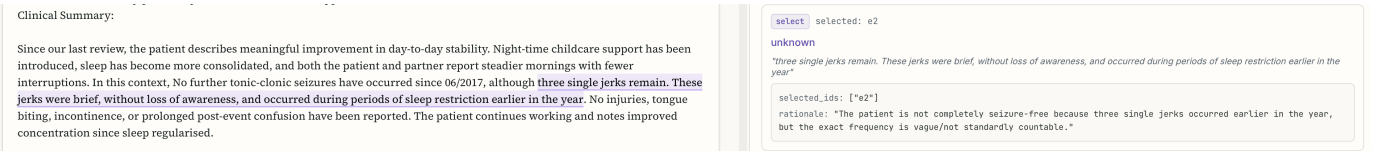


Fig. 4: Infrequent → unknown (development example). Gold-standard label: 3 per 14 months. Model selected unknown.

TABLE VII: Descriptive inventory output on 100 dev750 synthetic letters.

Family	Letters ≥ 1	Facts	Median (range)	Common named types
Diagnosis	76	130	1 (0–4)	epilepsy, focal epilepsy, generalised epilepsy, secondary generalised seizures, generalised seizures
Prescription	81	204	2 (0–9)	levetiracetam, lamotrigine, sodium valproate, clobazam, topiramate
Investigations	40	68	0 (0–3)	MRI normal, EEG abnormal, EEG normal, MRI abnormal
Any reported family	93	402	4 (0–11)	—

The local open-weight models used in the reported extracts were served with native Ollama chat at <http://localhost:11434>. In particular, Qwen 3.8 27B and Gemma 4 26B used the `ollama_chat/...` route.

C. Local web app

A local web application was built to inspect letters and pipeline records side by side as they moved through find, encode and select (Appendix C). It was used during development to examine candidate events, quoted evidence, rule rewrites and the final label, and to carry out qualitative error analysis on the development set. Held-out letters were not opened in this interface.

The interface is a Next.js 16 and React 19 application styled with Tailwind CSS. TanStack Query loads replayed pipeline records from the local Python API. The assessed repository includes this frontend alongside the extraction code.

D. Verification, testing, and reproducibility

Pytest provided automated checks for the extraction pipeline and its data handling. Ruff checked Python style and common implementation errors, while mypy checked the consistency of annotated interfaces. Together these tools supported the implementation and testing stages by detecting regressions before experiments were run.

Reproducibility was supported by the dependency declaration and lockfile, prespecified model routes and request settings, and per-cell experiment artifacts retaining the prompt or program version, split, scorer, and retry settings.

The assessed code repository is on QUB EEECS GitLab at https://gitlab.eecs.qub.ac.uk/40466218/clinical_extraction.git.

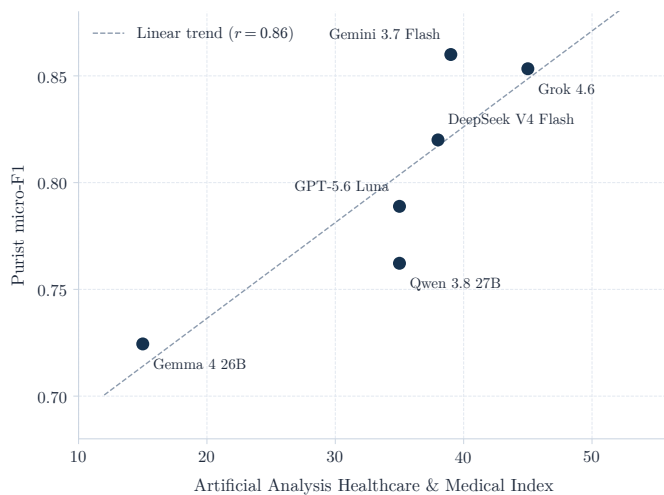


Fig. 5: Hybrid Purist micro-F1 on Gan test450 versus a published healthcare capability index for each extraction model. Higher index values track higher Purist stops on this synthetic task; the association is descriptive, not causal.

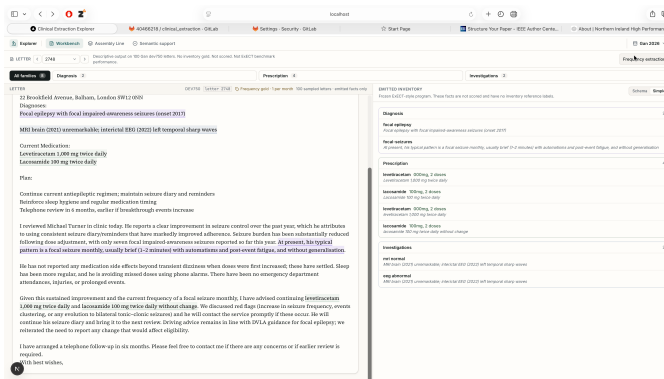


Fig. 6: Example of extracting multiple entities from the letter in a broader clinical inventory using the ExECT schema.

IX. CRITICAL APPRAISAL

A. Successes

The project produced a reproducible pipeline for extracting a single current seizure-frequency label from synthetic epilepsy clinic letters. Its main contribution was to separate the task into two inspectable stages: an LLM extracted candidate evidence into a structured record, and a decision stage returned the final label. Holding the extraction record fixed made it possible to compare a rules-based system with a second LLM call. Empirical analysis demonstrated that the Hybrid method performed better under all conditions tested, particularly for smaller models more likely to be deployed on hospital servers.

The pipeline showed consistent performance on synthetic development (0.83 / 0.77), synthetic test (0.82 / 0.77) and the external set of 300 real patient letters (0.80 / 0.77), demonstrating that both Hybrid and LLM-only transfer. The original research objective was an empirical study of reliability, and this is the strongest signal of reliability.

The most important design decision was to narrow the dissertation to the seizure frequency task. The original project explored both Gan and ExECTv2, but the two datasets have different annotation policies, outcome definitions, scorers, and error profiles. Presenting both as equal dissertation results would have required the reader to work too hard. Narrowing the results produced a cleaner argument and provided more room to demonstrate the methodology.

B. Limitations and challenges

The main limitation is the primary data source used. A single model (DeepSeek) was run on an external test set of patient data and showed strong transfer, but all other evaluations in the research used synthetic data. Synthetic data has been shown to produce more positive results in prior papers as it contains fewer typos, idiosyncrasies and a more consistent structure than real data. Past results have also shown that performance on one corpus of real patient data is not evidence that it will translate to other hospital templates or clinician writing styles. Thus, the results are promising but must be interpreted within those boundaries.

The secondary limitation is the lack of evidence that the method transfers to other tasks. The ExECTv2 work was designed to illustrate that the same methods can be applied to a task that requires a much richer schema: multiple clinical categories, multiple entities per category and multiple attributes per entity, vs. the single-label extraction task demonstrated here. That must be demonstrated in future work to show the two-stage extraction pipeline architecture generalises beyond a very specific task design.

C. Lessons learned and future work

The main lesson learned is that there is no substitute for examining the data in depth. Early experiments were run with a focus on the high-level metrics: precision, recall, and the F1 score. Iterations of prompt engineering and rule design would improve performance in one area and harm it in another. When I started to look in depth at the 400-word clinic letters themselves, the predictions the model made and, most importantly, the gold-standard annotations, it became clear that the complicated part of the task was a highly specific decision policy to choose the right answer. However, I avoided doing that early on because the letters were long, dense and the annotations were on a subject I lacked domain expertise in. Building a deeper understanding of what the data represented made the architecture design and optimisation process much simpler.

The next step is to evaluate whether the pipeline has clinical value. This should include expert review of whether the cited evidence supports the selected label, and prospective assessment of how clinicians would use or contest the outputs. The independent real-letter scores already show that the package can be run and scored outside the development environment; they do not replace those studies. Those studies would need their own data-access, ethics, and validation decisions. Until

then, the system can only be presented as a reproducible research prototype rather than a clinical decision tool.

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APPENDIX A

LLM PROMPT USED IN THE EXTRACT STAGE

```

{
  "task": "Read the clinical note. Extract seizure-frequency facts as slim events, then select the current burden.",
  "instructions": [
    "Read the full clinical note and extract source-near seizure-frequency facts.",
    "Return events as slim clinical facts, not fully normalized answer records. Use raw_value for the text's stated rate, duration, last-event statement, or unknown/no-reference cue.",
    "Event kind must be one of frequency_rate, cluster_frequency, seizure_free, last_event_only, unknown_frequency, or no_reference.",
    "Use one no_reference event only when the note contains no usable seizure-frequency evidence. Do not use no_reference when seizures are discussed but frequency is unclear; use unknown_frequency instead.",
    "Keep seizure-free statements separate from unknown or last-event-only statements. Do not select seizure-free if other current seizure-like events remain active.",
    "Selection must choose the highest current or recent seizure burden across semiologies when several current seizure types are present.",
    "If the note gives an overall current seizure count plus a breakdown by seizure type, select the overall count for the seizure-frequency label rather than only the clinically most severe subtype count.",
    "Write the seizure-frequency label using only the allowed forms. Copy an example and change the numbers if needed.",
    "If the selected event has a countable stated rate, put the note's wording in raw_value and write the seizure-frequency label from the allowed forms.",
    "When the note says a last event occurred on a date and the patient has been well, stable, or seizure-free since, still extract the dated last-event fact as its own event even if the selection is seizure-free.",
    "When the note says a count such as 3 or 4 jerks occurred since a dated last tonic-clonic seizure, keep the source count and the dated anchor available in the event list.",
    "Every evidence value must be an exact substring from the note when possible.",
    "Return exactly one JSON object with no markdown."
  ],
  "label_forms": {
    "rules": [
      "Write the label using only the forms below. Copy an example and change the numbers if needed.",
      "Use digits, not word numbers.",
      "Use day, week, month, or year. Prefer the singular word.",
      "A night count is a day count: once per night becomes 1 per day.",
      "Flatten a bound: at most four per day, or ≤ four per day, becomes 4 per day.",
      "Do not write ≤, ≥, up to, at most, several, handful, frequent, or a calendar date.",
      "If none of these forms fit, write unknown when seizures are discussed, or no seizure frequency reference when there is no usable frequency evidence."
    ],
    "forms": [
      {
        "form": "N per unit",
        "description": "A single count in one day, week, month, or year.",
        "examples": [
          "1 per day",
          "4 per week",
          "3 per month",
          "2 per year"
        ]
      },
      {
        "form": "N per N unit",
        "description": "A single count over a stated number of days, weeks, months, or years.",
        "examples": [
          "1 per 2 day",
          "1 per 2 week",
          "2 per 6 month"
        ]
      },
      {
        "form": "N to N per unit",
        "description": "A count range in one day, week, month, or year.",
        "examples": [
          "2 to 3 per day",
          "2 to 4 per week",
          "3 to 5 per month",
          "2 to 4 per year"
        ]
      },
      {
        "form": "N to N per N unit",
        "description": "A count range over a stated number of days, weeks, months, or years.",
        "examples": [
          "2 to 4 per 3 month",
          "2 to 3 per 2 week"
        ]
      },
      {
        "form": "multiple per unit",
        "description": "More than one seizure in one day, week, or month, with no number given.",
        "examples": [
          "multiple per day",
          "multiple per week",
          "multiple per month"
        ]
      },
      {
        "form": "multiple per N unit",
        "description": "More than one seizure over a stated number of months, with no number given.",
        "examples": [
          "multiple per 2 month"
        ]
      },
      {
        "form": "N per multiple unit",
        "description": "A count over an unstated number of days or months.",
        "examples": [
          "1 per multiple day",
          "1 per multiple month"
        ]
      },
      {
        "form": "cluster per unit, N per cluster",
        "description": "How often clusters happen, and how many seizures are in each cluster. The cluster count or the time between clusters may be a range.",
        "examples": [
          "1 cluster per day, 5 per cluster",
          "1 cluster per week, 4 per cluster",
          "1 cluster per 4 month, 5 per cluster",
          "6 cluster per month, 4 per cluster",
          "1 cluster per 4 to 5 day, 2 per cluster"
        ]
      },
      {
        "form": "cluster per unit, range per cluster",
        "description": "How often clusters happen, and a range of seizures in each cluster.",
        "examples": [
          "1 cluster per day, 2 to 4 per cluster",
          "1 cluster per week, 3 to 6 per cluster",
          "1 cluster per month, 4 to 6 per cluster"
        ]
      }
    ]
  }
}

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```

    ],
    {
      "form": "cluster per unit, multiple per cluster",
      "description": "How often clusters happen, and more than one seizure in each cluster with no number given.",
      "examples": [
        "1 cluster per day, multiple per cluster",
        "1 cluster per week, multiple per cluster",
        "1 cluster per month, multiple per cluster"
      ]
    },
    {
      "form": "unknown cluster count",
      "description": "Clusters are described, but how often they happen is not known.",
      "examples": [
        "unknown, 5 per cluster",
        "unknown, 2 to 4 per cluster",
        "unknown, multiple per cluster"
      ]
    },
    {
      "form": "seizure free for a duration",
      "description": "No seizures for a stated length of time.",
      "examples": [
        "seizure free for 6 month",
        "seizure free for 2 year",
        "seizure free for 3 months",
        "seizure free for 2 years",
        "seizure free for 2 to 3 months"
      ]
    },
    {
      "form": "seizure free for a vague duration",
      "description": "No seizures for an unstated length of time.",
      "examples": [
        "seizure free for multiple month",
        "seizure free for multiple year"
      ]
    },
    {
      "form": "unknown",
      "description": "Seizures are discussed, but there is no usable frequency.",
      "examples": [
        "unknown"
      ]
    },
    {
      "form": "no seizure frequency reference",
      "description": "There is no usable frequency evidence.",
      "examples": [
        "no seizure frequency reference"
      ]
    }
  ],
  "event_schema": {
    "event_id": "stable string such as e1",
    "kind": [
      "frequency_rate",
      "cluster_frequency",
      "seizure_free",
      "last_event_only",
      "unknown_frequency",
      "no_reference"
    ],
    "raw_value": "source-near expression or null",
    "applies_to": "seizure type or clinical target, or null",
    "time_window": "source-near current/recent/historical window, or null",
    "temporality": [
      "current",
      "recent",
      "historical",
      "future",
      "unclear"
    ]
  },

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  ],
  "assertion_status": [
    "asserted",
    "negated",
    "historical",
    "hypothetical",
    "unknown"
  ],
  "evidence": "exact note substring",
  "notes": "optional short note or null"
},
"selection_schema": {
  "selected_event_ids": "list of selected event_id strings",
  "final_kind": [
    "frequency",
    "seizure_free",
    "unknown",
    "no_reference",
    "unresolved_multiple"
  ],
  "final_label": "normalized label, or null if not directly countable",
  "evidence": "exact note substring supporting the final selection",
  "confidence": [
    "low",
    "medium",
    "high"
  ],
  "rationale": "brief clinical reason for selecting these events"
}
}

```

APPENDIX B

LLM PROMPT USED IN THE DECIDE STAGE

```

{
  "task": "Choose which events describe how often seizures are happening now.",
  "instructions": [
    "Each event has a seizure-frequency label and a supporting quote. Use both. The quote can justify a different answer than the label. Use only facts already in the events or their quotes. Any new label must match one of the label forms.",
    "A first choice is given. Keep it unless one of the cases below applies.",
    "Return the selected event ids. Write a new label only when no single event is the answer. Do not add events. Do not write a new quote. Return exactly one JSON object with no markdown."
  ],
  "cases": [
    {
      "title": "Usual gap",
      "instruction": "The first choice is unknown, 1 per day, or multiple per day during short or occasional daily spells. Another event gives the usual gap between seizures. Prefer that usual gap.",
      "example": {
        "first_choice": {
          "selected_event_ids": [
            "e1"
          ],
          "label": "1 per day"
        },
        "events": [
          {
            "event_id": "e1",
            "label": "1 per day",
            "kind": "frequency_rate",
            "temporality": "current",
            "assertion_status": "asserted",
            "applies_to": "seizures",
            "time_window": null,
            "evidence": "brief periods of daily seizures"
          },
          {
            "event_id": "e2",
            "label": "1 per 2 week",

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        "kind": "frequency_rate",
        "temporality": "current",
        "assertion_status": "asserted",
        "applies_to": "seizures",
        "time_window": null,
        "evidence": "usually every 2 weeks"
    }
},
"answer": {
    "selected_event_ids": [
        "e2"
    ]
}
},
{
    "title": "Usual rate, not a year total",
    "instruction": "The first choice is how many seizures
so far this year. Another event gives the usual or
typical rate. Prefer the usual rate.",
    "example": {
        "first_choice": {
            "selected_event_ids": [
                "e1"
            ],
            "label": "4 per year"
        },
        "events": [
            {
                "event_id": "e1",
                "label": "4 per year",
                "kind": "frequency_rate",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "seizures",
                "time_window": "this year",
                "evidence": "4 seizures so far this year"
            },
            {
                "event_id": "e2",
                "label": "1 per month",
                "kind": "frequency_rate",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "seizures",
                "time_window": null,
                "evidence": "typically 1 per month"
            }
        ],
        "answer": {
            "selected_event_ids": [
                "e2"
            ]
        }
    }
},
{
    "title": "Recent seizures after a quiet spell",
    "instruction": "The first choice is unknown or no
seizure frequency reference. The events give a recent
count and how long the person had been seizure-free.
Write that count over that time. Do this only if no
event already has that label.",
    "example": {
        "first_choice": {
            "selected_event_ids": [
                "e1"
            ],
            "label": "unknown"
        },
        "events": [
            {
                "event_id": "e1",
                "label": "unknown",
                "kind": "unknown_frequency",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "tonic-clonic seizures",
                "time_window": null,
                "evidence": "2 tonic-clonic seizures"
            },
            {
                "event_id": "e2",

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        "label": "seizure free for 6 month",
        "kind": "seizure_free",
        "temporality": "current",
        "assertion_status": "asserted",
        "applies_to": "seizures",
        "time_window": "6 months",
        "evidence": "seizure-free for 6 months before
that"
    }
},
"answer": {
    "selected_event_ids": [
        "e1",
        "e2"
    ],
    "label": "2 per 6 month"
}
},
{
    "title": "Not epileptic seizures",
    "instruction": "The first choice is unknown or no
seizure frequency reference. The current events are
attacks that are not epileptic seizures. Write seizure
free for multiple year.",
    "example": {
        "first_choice": {
            "selected_event_ids": [
                "e1"
            ],
            "label": "unknown"
        },
        "events": [
            {
                "event_id": "e1",
                "label": "unknown",
                "kind": "unknown_frequency",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "non-epileptic attacks",
                "time_window": null,
                "evidence": "these are non-epileptic attacks"
            },
            {
                "event_id": "e2",
                "label": "unknown",
                "kind": "unknown_frequency",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "seizures",
                "time_window": null,
                "evidence": "no epileptic seizures"
            }
        ],
        "answer": {
            "selected_event_ids": [
                "e1",
                "e2"
            ],
            "label": "seizure free for multiple year"
        }
    }
},
{
    "title": "Month counts",
    "instruction": "Quotes name how many seizures happened
in named months, such as 3 in March and 6 in May. Add
those counts and write the total over that time. Use
this instead of the last month alone, unknown, or a
short or vague seizure-free label. Do not replace a
rate per day or per week. Do not replace seizure-free
for 4 months or longer, or for any number of years. You
may replace a vague seizure free for multiple label.",
    "example": {
        "first_choice": {
            "selected_event_ids": [
                "e2"
            ],
            "label": "6 per month"
        },
        "events": [
            {
                "event_id": "e1",

```

```

        "label": "3 per month",
        "kind": "frequency_rate",
        "temporality": "current",
        "assertion_status": "asserted",
        "applies_to": "seizures",
        "time_window": "March",
        "evidence": "3 seizures in March"
    },
    {
        "event_id": "e2",
        "label": "6 per month",
        "kind": "frequency_rate",
        "temporality": "current",
        "assertion_status": "asserted",
        "applies_to": "seizures",
        "time_window": "May",
        "evidence": "6 seizures in May"
    }
],
"answer": {
    "selected_event_ids": [
        "e1",
        "e2"
    ],
    "label": "9 per 3 month"
}
},
{
    "title": "Dated seizures",
    "instruction": "Two or more events name seizures on different dates or months, or a quote already says how many seizures happened in a number of months. Write that count over that time. Use this instead of unknown, seizure-free after the last date, or a monthly or yearly rate when that time is more than 1 month. Do not replace a rate per day or per week. Do this only if the last dated event looks current or recent. Ignore dates described as before an improvement.",
    "example": {
        "first_choice": {
            "selected_event_ids": [
                "e1"
            ],
            "label": "unknown"
        },
        "events": [
            {
                "event_id": "e1",
                "label": "unknown",
                "kind": "unknown_frequency",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "seizures",
                "time_window": "March 2019",
                "evidence": "a seizure in March 2019"
            },
            {
                "event_id": "e2",
                "label": "unknown",
                "kind": "unknown_frequency",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "seizures",
                "time_window": "May 2019",
                "evidence": "a seizure in May 2019"
            }
        ],
        "answer": {
            "selected_event_ids": [
                "e1",
                "e2"
            ],
            "label": "2 per 2 month"
        }
    },
    {
        "title": "Burst after a change",
        "instruction": "The first choice is seizure-free, or a rate per day or per week. The quotes describe seizures soon after a treatment change, then no further seizures for a stated time. Write that burst count over

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the quiet time. Do this only if the quotes already give both the burst and the quiet time.",
        "example": {
            "first_choice": {
                "selected_event_ids": [
                    "e2"
                ],
                "label": "seizure free for 6 month"
            },
            "events": [
                {
                    "event_id": "e1",
                    "label": "3 per week",
                    "kind": "frequency_rate",
                    "temporality": "current",
                    "assertion_status": "asserted",
                    "applies_to": "seizures",
                    "time_window": "following week",
                    "evidence": "3 seizures shortly after the dose change"
                },
                {
                    "event_id": "e2",
                    "label": "seizure free for 6 month",
                    "kind": "seizure_free",
                    "temporality": "current",
                    "assertion_status": "asserted",
                    "applies_to": "seizures",
                    "time_window": "6 months",
                    "evidence": "no further seizures since then"
                }
            ],
            "answer": {
                "selected_event_ids": [
                    "e1",
                    "e2"
                ],
                "label": "3 per 6 month"
            }
        },
        {
            "title": "Short quiet spell after a last event",
            "instruction": "The first choice is seizure-free for weeks, or for fewer than 6 months. A quote names the last seizure on a calendar day, or a burst count, and says the person has been well since then. Write that count over that time. If the quiet spell is 5 weeks or less, write the count per month.",
            "example": {
                "first_choice": {
                    "selected_event_ids": [
                        "e2"
                    ],
                    "label": "seizure free for 3 week"
                },
                "events": [
                    {
                        "event_id": "e1",
                        "label": "unknown",
                        "kind": "last_event_only",
                        "temporality": "current",
                        "assertion_status": "asserted",
                        "applies_to": "seizures",
                        "time_window": "12 March",
                        "evidence": "last seizure on 12 March"
                    },
                    {
                        "event_id": "e2",
                        "label": "seizure free for 3 week",
                        "kind": "seizure_free",
                        "temporality": "current",
                        "assertion_status": "asserted",
                        "applies_to": "seizures",
                        "time_window": "3 weeks",
                        "evidence": "remained well since then"
                    }
                ],
                "answer": {
                    "selected_event_ids": [
                        "e1",
                        "e2"
                    ],

```

```

        "label": "1 per month"
    }
},
{
    "title": "Overall count",
    "instruction": "If several seizure types are listed,
choose the highest current or recent overall count, not
a breakdown by type.",
    "example": {
        "first_choice": {
            "selected_event_ids": [
                "e1"
            ],
            "label": "4 per month"
        },
        "events": [
            {
                "event_id": "e1",
                "label": "4 per month",
                "kind": "frequency_rate",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "focal seizures",
                "time_window": "this month",
                "evidence": "4 focal seizures this month"
            },
            {
                "event_id": "e2",
                "label": "2 per month",
                "kind": "frequency_rate",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "tonic-clonic seizures",
                "time_window": "this month",
                "evidence": "2 tonic-clonic seizures this month"
            },
            {
                "event_id": "e3",
                "label": "6 per month",
                "kind": "frequency_rate",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "seizures",
                "time_window": "this month",
                "evidence": "6 seizures this month in total"
            }
        ],
        "answer": {
            "selected_event_ids": [
                "e3"
            ]
        }
    }
},
{
    "title": "Do not choose seizure-free while events
continue",
    "instruction": "Do not choose seizure-free if other
current seizure-like events are still happening.",
    "example": {
        "first_choice": {
            "selected_event_ids": [
                "e1"
            ],
            "label": "seizure free for 2 month"
        },
        "events": [
            {
                "event_id": "e1",
                "label": "seizure free for 2 month",
                "kind": "seizure_free",
                "temporality": "current",
                "assertion_status": "asserted",
                "applies_to": "seizures",
                "time_window": "2 months",
                "evidence": "seizure free for 2 months"
            },
            {
                "event_id": "e2",
                "label": "1 per week",
                "kind": "frequency_rate",
                "temporality": "current",

```

```

        "assertion_status": "asserted",
        "applies_to": "absences",
        "time_window": null,
        "evidence": "absences continue weekly"
    }
},
    "answer": {
        "selected_event_ids": [
            "e2"
        ]
    }
},
    "label_forms": {
        "rules": [
            "Write the label using only the forms below. Copy an
example and change the numbers if needed.",
            "Use digits, not word numbers.",
            "Use day, week, month, or year. Prefer the singular
word.",
            "A night count is a day count: once per night becomes
1 per day.",
            "Flatten a bound: at most four per day, or ≤ four per
day, becomes 4 per day.",
            "Do not write <=, >=, up to, at most, several, handful
, frequent, or a calendar date.",
            "If none of these forms fit, write unknown when
seizures are discussed, or no seizure frequency
reference when there is no usable frequency evidence."
        ],
        "forms": [
            {
                "form": "N per unit",
                "description": "A single count in one day, week,
month, or year.",
                "examples": [
                    "1 per day",
                    "4 per week",
                    "3 per month",
                    "2 per year"
                ]
            },
            {
                "form": "N per N unit",
                "description": "A single count over a stated number
of days, weeks, months, or years.",
                "examples": [
                    "1 per 2 day",
                    "1 per 2 week",
                    "2 per 6 month"
                ]
            },
            {
                "form": "N per N to N unit",
                "description": "A single count over a range of days,
weeks, months, or years.",
                "examples": [
                    "1 per 2 to 3 day",
                    "1 per 2 to 3 week",
                    "1 per 4 to 6 month"
                ]
            },
            {
                "form": "N to N per unit",
                "description": "A count range in one day, week,
month, or year.",
                "examples": [
                    "2 to 3 per day",
                    "2 to 4 per week",
                    "3 to 5 per month",
                    "2 to 4 per year"
                ]
            },
            {
                "form": "N to N per N unit",
                "description": "A count range over a stated number
of days, weeks, months, or years.",
                "examples": [
                    "2 to 4 per 3 month",
                    "2 to 3 per 2 week"
                ]
            }
        ],
    },

```

```

{
  "form": "multiple per unit",
  "description": "More than one seizure in one day,
week, or month, with no number given.",
  "examples": [
    "multiple per day",
    "multiple per week",
    "multiple per month"
  ]
},
{
  "form": "multiple per N unit",
  "description": "More than one seizure over a stated
number of months, with no number given.",
  "examples": [
    "multiple per 2 month"
  ]
},
{
  "form": "N per multiple unit",
  "description": "A count over an unstated number of
days or months.",
  "examples": [
    "1 per multiple day",
    "1 per multiple month"
  ]
},
{
  "form": "cluster per unit, N per cluster",
  "description": "How often clusters happen, and how
many seizures are in each cluster. The cluster count or
the time between clusters may be a range.",
  "examples": [
    "1 cluster per day, 5 per cluster",
    "1 cluster per week, 4 per cluster",
    "1 cluster per 4 month, 5 per cluster",
    "6 cluster per month, 4 per cluster",
    "1 cluster per 4 to 5 day, 2 per cluster"
  ]
},
{
  "form": "cluster per unit, range per cluster",
  "description": "How often clusters happen, and a
range of seizures in each cluster.",
  "examples": [
    "1 cluster per day, 2 to 4 per cluster",
    "1 cluster per week, 3 to 6 per cluster",
    "1 cluster per month, 4 to 6 per cluster"
  ]
},
{
  "form": "cluster per unit, multiple per cluster",
  "description": "How often clusters happen, and more
than one seizure in each cluster with no number given
.",
  "examples": [
    "1 cluster per day, multiple per cluster",
    "1 cluster per week, multiple per cluster",
    "1 cluster per month, multiple per cluster"
  ]
},
{
  "form": "unknown cluster count",
  "description": "Clusters are described, but how
often they happen is not known.",
  "examples": [
    "unknown, 5 per cluster",
    "unknown, 2 to 4 per cluster",
    "unknown, multiple per cluster"
  ]
},
{
  "form": "seizure free for a duration",
  "description": "No seizures for a stated length of
time.",
  "examples": [
    "seizure free for 6 month",
    "seizure free for 2 year",
    "seizure free for 3 months",
    "seizure free for 2 years",
    "seizure free for 2 to 3 months"
  ]
},

```

```

{
  "form": "seizure free for a vague duration",
  "description": "No seizures for an unstated length
of time.",
  "examples": [
    "seizure free for multiple month",
    "seizure free for multiple year"
  ]
},
{
  "form": "unknown",
  "description": "Seizures are discussed, but there is
no usable frequency.",
  "examples": [
    "unknown"
  ]
},
{
  "form": "no seizure frequency reference",
  "description": "There is no usable frequency
evidence.",
  "examples": [
    "no seizure frequency reference"
  ]
}
]
},
"selection_schema": {
  "selected_event_ids": "ids of the events used for the
answer",
  "label": "new seizure-frequency label from the label
forms, only when no single event is the answer;
otherwise omit"
}
}

```

APPENDIX C WEB APP DEMO

Find 2
2 candidates

Encode 2
2 events · 7 per 10 month

Select 1
1 per month · 1 selected

Score 1
Purist match · seizure_freq_1_per_mon

LETTER

Diagnoses:
Focal epilepsy with focal impaired-awareness seizures (onset 2017)

MRI brain (2021) unremarkable; interictal EEG (2022) left temporal sharp waves

Current Medication:
Levetiracetam 1,000 mg twice daily
Lacosamide 100 mg twice daily

Plan:

Continue current antiepileptic regimen; maintain seizure diary and reminders
Reinforce sleep hygiene and regular medication timing
Telephone review in 6 months, earlier if breakthrough events increase

I reviewed Michael Turner in clinic today. He reports a clear improvement in seizure control over the past year, which he attributes to using consistent seizure diary/reminders that have markedly improved adherence. Seizure burden has been substantially reduced following dose adjustment, with only seven focal impaired-awareness seizures reported so far this year. At present, his typical pattern is a focal seizure monthly, usually brief (1-2 minutes) with automatisms and post-event fatigue, and without generalisation.

He has not reported any medication side effects beyond transient dizziness when doses were first increased; these have settled. Sleep has been more regular, and he is avoiding missed doses using phone alarms. There have been no emergency department attendances, injuries, or prolonged events.

Given this sustained improvement and the current frequency of a focal seizure monthly, I have advised continuing levetiracetam 1,000 mg twice daily and lacosamide 100 mg twice daily without change. We discussed red flags (increase in seizure frequency, events clustering, or any evolution to bilateral tonic-clonic seizures) and he will contact the service promptly if these occur. He will continue his seizure diary and bring it to the next review. Driving advice remains in line with DVLA guidance for focal epilepsy; we reiterated the need to report any change that would affect eligibility.

I have arranged a telephone follow-up in six months. Please feel free to contact me if there are any concerns or if earlier review is required.

With best wishes,

VALIDATION Letter: 2748 1 per month 2,359 chars

Find Raw candidate events found in the note by find-stage rules or LLM claims.
only seven focal impaired-awareness seizures reported so far this yeare1 frequency_rate
seven focal impaired-awareness seizures reported so far this year
"only seven focal impaired-awareness seizures reported so far this year"
temporality: "recent"
assertion_status: "asserted"
applies_to: "focal impaired-awareness seizures"
notes: "year-to-date count"e2 frequency_rate
a focal seizure monthly
"At present, his typical pattern is a focal seizure monthly"
temporality: "current"
assertion_status: "asserted"
applies_to: "focal seizure"
notes: "current baseline frequency"

(a) Find: two candidate frequency statements in the letter.

Find 2
2 candidates

Encode 2
2 events · 7 per 10 month

Select 1
1 per month · 1 selected

Score 1
Purist match · seizure_freq_1_per_mon

LETTER

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I have arranged a telephone follow-up in six months. Please feel free to contact me if there are any concerns or if earlier review is required.

With best wishes,

VALIDATION Letter: 2748 1 per month 2,359 chars

Encode Candidates converted to codebook labels with semantic kinds and monthly frequencies.
only seven focal impaired-awareness seizures reported so far this yeare1 frequency_rate
7 per 10 month
"only seven focal impaired-awareness seizures reported so far this year"
original_label: "seven focal impaired-awareness seizures reported so far this year"
monthly_frequency: 0.7
rule: per_year_to_monthly_rate
rule_description: yearly frequency ÷ 12 months/yeare2 frequency_rate
1 per month
"At present, his typical pattern is a focal seizure monthly"
original_label: "a focal seizure monthly"
monthly_frequency: 1.0
rule: per_month_rate
rule_description: monthly frequency rate (1:1)

(b) Encode: each candidate written as a monthly rate.

Find 2
2 candidates

Encode 2
2 events · 7 per 10 month

Select 1
1 per month · 1 selected

Score 1
Purist match · seizure_freq_1_per_mon

LETTER

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He has not reported any medication side effects beyond transient dizziness when doses were first increased; these have settled. Sleep has been more regular, and he is avoiding missed doses using phone alarms. There have been no emergency department attendances, injuries, or prolonged events.

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VALIDATION Letter: 2748 1 per month 2,359 chars

Select The pipeline chooses a single final label from the encoded candidates.
At present, his typical pattern is a focal seizure monthlyselect selected: e2
1 per month
"At present, his typical pattern is a focal seizure monthly"
selected_ids: ["e2"]
rationale: "The note states the patient's current typical pattern is a focal seizure monthly."